Year 8

Mini Test 3

Geometry- Congruence of triangles and properties of quadrilaterals

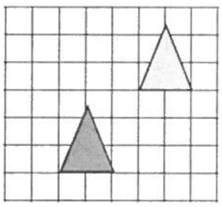
Name: ______________________________ Score: _____/50

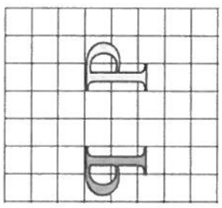
Show working and answers on this sheet. Show **ALL** working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

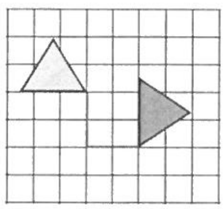
Time Allocated 50 minutes
Resources Permitted- Calculators and 1 A4 Page of notes.

Question 1 **[3, 1, 2, 2 = 8]**

a) Describe the transformation (**reflection, translation, or rotation**) to produce the congruent shape from the original:

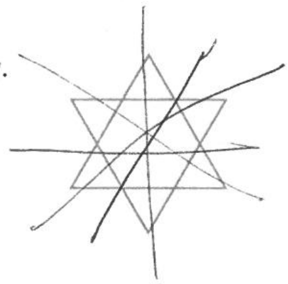
i) 

ii) 

iii) 

translation ✓ reflection ✓ rotation. ✓

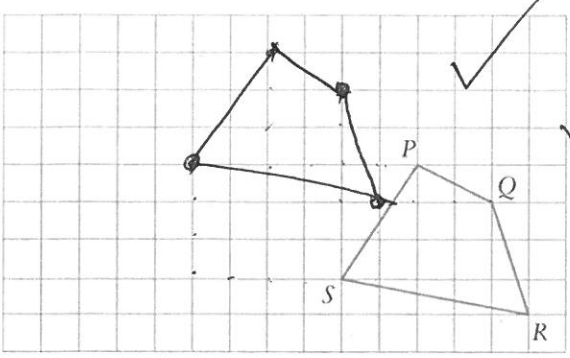
b Draw in all of the lines of symmetry.



✓

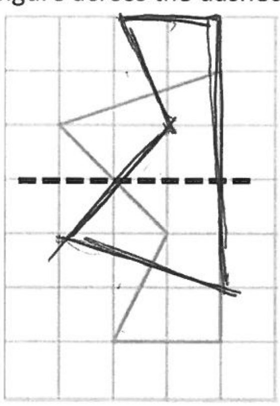
6 lines

c Translate *PQRS* 4 units left and 3 units up.



✓ ✓

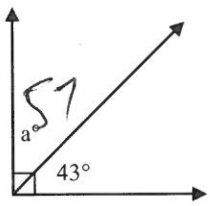
d Reflect the figure across the dashed line.

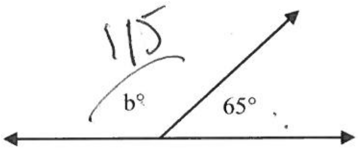


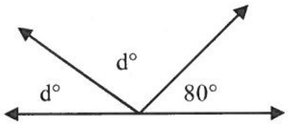
✓✓

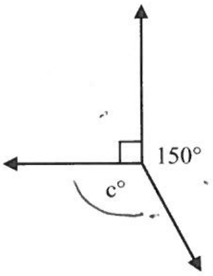
Question 2 **[8, 2 = 10]**

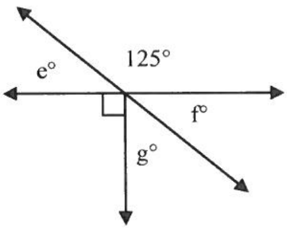
a For each of the following diagrams determine the value of the pronumeral

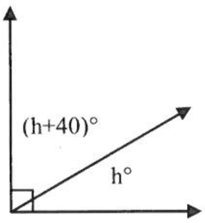












The two angles are such that one is 40° bigger than the other

a° 57

b° 115

c° 120

d° 50

e° 55

f° 55

g° 35

h° 25

$h + 40 + h = 90$

b For each of the following state whether the angles are complementary or supplementary.

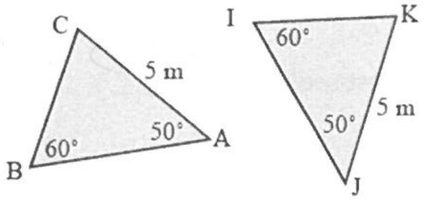
(i) 67° and 23° complementary

(ii) 5° and 175° Supplementary

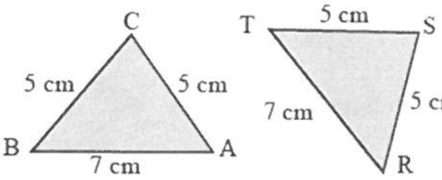
Question 3 **[2, 2, 2, 2= 8]**

For each of these pairs of congruent triangles write a congruency statement and indicate which congruency test used

i)



ii)

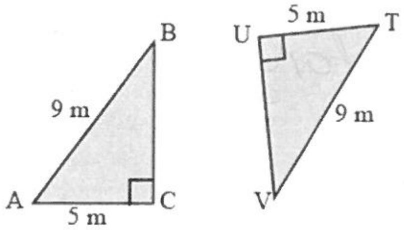


$\triangle ABC \equiv \triangle JIK$ ✓ $\triangle BCA \equiv \triangle TSR$ ✓

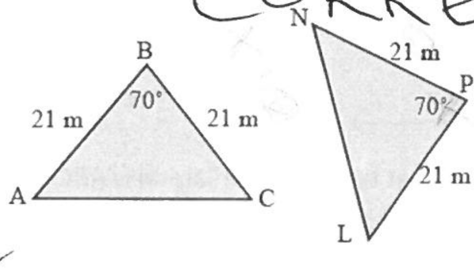
(AAS) ✓ (SSS). ✓

ORDER MUST BE CORRESPONDING

iii)



iv)



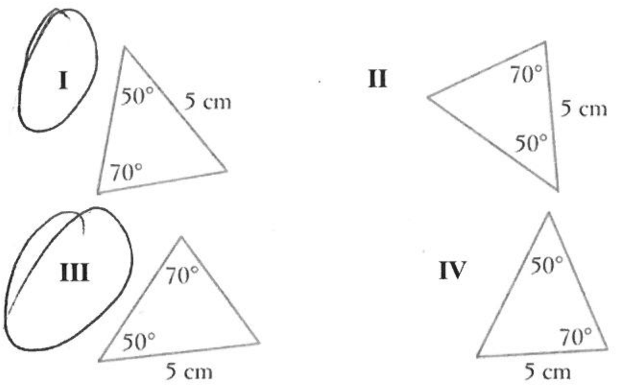
✓

$\triangle ABC \equiv \triangle TVU$ $\triangle ABC \equiv \triangle NPL$.

(RHS) ✓ (SAS). ✓

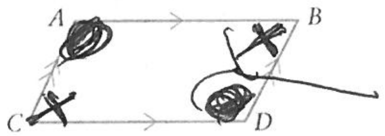
Question 4 **[1, 1 = 2]**

Of the four triangles, two of them are congruent. **Circle** the two that are congruent and state what test is used to prove it.



Test used AAS.

Question 5 **[1, 2, 1= 4]**



supplementary.

a What type of quadrilateral is *ABCD*?

parallelogram ✓

b Name a pair of equal angles.

$\angle CAB = \angle BDC$

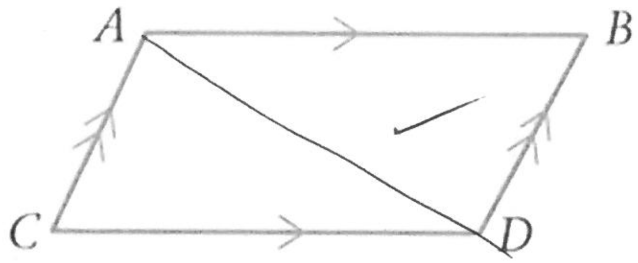
1 mark if they don't use 3 letters.

c Show two angles on the diagram that are supplementary

B & D

or A&C etc.

Question 6 **[1, 3, 2 = 6]**



a On the above figure construct diagonal AD.

b Complete the statements.

CD = AB (corresponding side)

AC= DB (corresponding side)

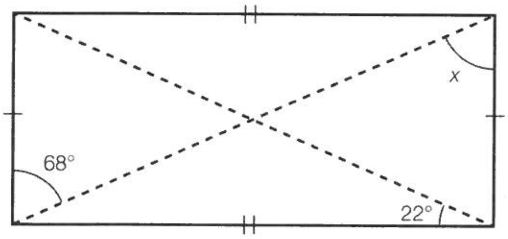
AD is common to both triangles.

c Now complete the Congruency Statement giving your reason.

$\triangle ACD \equiv \triangle DBA$ (SSS)

Question 7 **[3]**

Use congruent triangles and known angle facts to find the value of the pronumeral in the following rectangle. Give reasons for your answers.



~~ASA~~

$x = 68$ (Alternate angles are equal)

or other factual reason showing other angles.

Question 8 **[2, 2 = 4]**

Find the value of the unknown angle in each of the following.

a A quadrilateral with angles of 78°, 125°, 62° and *x*.

360 − 265 = 95 ✓ ✓

b A kite with a top angle of 65°, a base angle of 72°.

65 + 72 = 137

360 − 137 = 223 ✓

So each angle is 111·5° ✓

Question 9 **[3, 1, 1 = 5]**

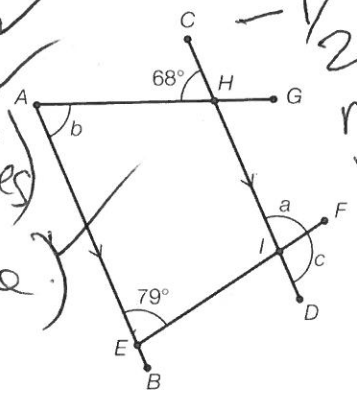
a Find the value of the pronumerals in the following quadrilateral. Give reasons for your answers.

b = 68 (Alternate angles) ✓

a = 79° (corresponding angles) ✓

c = 101 (CD straight line)

−1 each ½ for no reason.



b What other angles are equal to ∠*FID* (or *c*)?

∠FEB ✓

c What is the size of ∠*AHI*?

180 − 68 = 112 ✓